

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

**COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07**

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:25

Annexure A

Comparative Analysis

Code of Conduct:

I J. Kirthiga / 192421265 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. 20/20

I understand that any violation of academic integrity rules will result in disciplinary action.

J. Kirthiga
Signature of the student:

1. Comparison of Star and Bus Topologies

Star topology uses a central hub or switch and independent cables for each node, making it more expensive and reliable than a bus topology, which uses a single shared backbone cable that is cheap to install but prone to total network failure if cut.

Cost and Setup

- Star: Needs a central device (switch or hub) and separate wires for every single computer, raising costs.
- Bus: Uses one long main cable to link all devices in a line, lowering setup costs

Reliability and Faults

- Star: If one workstation cable breaks, only that single device goes offline.
- Bus: If the main backbone cable gets damaged anywhere, the entire network stops working

Speed and Traffic

- Star: Manages data flow well through the central node, lowering data crashes.
- Bus: Slows down as you add more devices because everyone shares the same line, leading to heavy traffic and data crashes.

Key Technical Criteria Comparison

- **Cable Media:** Star topology usually relies on Twisted Pair cables (like Cat6) paired with RJ-45 connectors. Bus topology traditionally depends on Coaxial cables using BNC connectors and terminators at both ends.
- **Data Flow Direction:** Star networks utilize Point-to-Point links between each device and the switch, allowing simultaneous data tracks. Bus networks utilize a Broadcast mechanism where data travels along the single line to every node until it finds the recipient.

- **Troubleshooting Ease:** Star setups make isolating faults incredibly fast because the switch can identify the exact port experiencing an error. Bus setups make pinpointing faults highly difficult, as a break anywhere breaks the signal reflection for the entire line.
- **Expansion Limitations:** Star expansion is limited only by the number of open ports on the central switch or hub. Bus expansion is strictly limited by the physical length of the cable and signal degradation (attenuation).

Advantages and Disadvantages

Star Topology

- Pros: Highly scalable, easy to configure, simple to troubleshoot, and safe from single-node failures.
- Cons: Complete dependency on the central device (if the switch dies, the whole network goes down) and high initial hardware costs.

Bus Topology

- Pros: Excellent for temporary or small setups, requires significantly less total cable length, and needs no active central hardware.
- Cons: Heavily prone to data collisions, difficult to reconfigure while online, and a single break creates total network downtime.

Feature	Star Topology	Bus Topology
Central Device	Required (Switch/Hub)	None
Cable Type	Twisted Pair (UTP/STP)	Coaxial
Single Cable Break	Impacts only one node	Down the entire network
Central Device Failure	Down the entire network	N/A
Installation Difficulty	Moderate	Very Easy
Data Collision Risk	Very Low (with a switch)	Very High

2. Mesh topology provides high fault tolerance because nodes connect to multiple devices, allowing data to find new paths if a link breaks. Star networks rely on a central hub, meaning the whole network stops if that hub fails. General guidance on mesh topology outlines these structural differences.

Main Differences

- **Mesh Design:** Every device links to several others, creating redundant paths and self-healing.
- **Star Design:** Devices connect only to a central hub or switch.
- **Failure Impact:** A single broken link in a mesh does not stop traffic, but a broken hub in a star stops everything.
- **Cost and Setup:** Mesh needs more cables and is costly; star is simpler and cheaper to build

Scalability in Mesh and Star Topologies

- Mesh topology is not easily scalable because adding a new node requires multiple connections to existing nodes, increasing complexity and cost significantly. In contrast, star topology is highly scalable, as new nodes can be easily added by connecting them to the central switch without affecting the rest of the network. This makes star topology more suitable for growing networks.

Installation and Maintenance Complexity

Mesh topology is complex to install and maintain due to the large number of connections between nodes. Troubleshooting can also be difficult because of multiple data paths. On the other hand, star topology is easier to install and manage since all connections are centralized, making it simpler to detect and fix faults.

Data Transmission Efficiency

In mesh topology, data can take the most efficient path due to multiple available routes, resulting in faster and more reliable communication.

However, managing these paths can be complex. In star topology, data transmission is efficient and organized because all communication passes through the central device, reducing confusion but depending heavily on the switch's performance.

Cost and Cabling Requirements

Mesh topology is very expensive due to the need for extensive cabling and multiple connections between devices. Each node must be connected to many others, increasing infrastructure costs. In contrast, star topology requires less cabling than mesh and is more cost-effective, although it is still more expensive than bus topology.

Security Considerations

Mesh topology provides better security because data can travel through multiple paths, making it harder to intercept all communications. Star topology also offers good security since traffic passes through a central device, allowing monitoring and control, but it can be vulnerable if the central device is compromised.

3. Installation Simplicity and Complexity in Bus and Mesh Topologies

Key Differences

- **Cabling:** Bus needs one main cable plus short drop lines; mesh needs a vast web of individual wires for every device pair.
- **Cost:** Bus is very cheap; mesh is very expensive due to hardware and port requirements.
- **Setup:** Bus is quick and simple; mesh is slow and hard to build.
- **Reliability:** Bus fails if the main line breaks; mesh stays up through alternate paths if a single link fails
- **Scalability in Bus and Mesh Topologies**

Bus topology has limited scalability because adding more devices increases traffic on the single backbone cable, leading to reduced performance. There is also a limit to the cable length and number of nodes that can be connected. In contrast, mesh topology is difficult to scale due to the need for multiple connections between nodes. Adding a new device requires connecting it to all existing nodes, making expansion complex and expensive.

- **Fault Detection and Troubleshooting**

In bus topology, identifying faults can be challenging because all devices share a single communication line. A problem in the backbone cable can affect the entire network, making troubleshooting time-consuming. On the other hand, mesh topology allows easier fault detection since multiple paths exist, and failures can often be isolated without affecting the entire network.

- **Network Performance Under Load**

Bus topology suffers from performance degradation as network traffic increases because all devices share the same communication medium. This leads to data collisions and delays. In contrast, mesh topology

performs efficiently even under heavy load because multiple paths allow simultaneous data transmission, reducing congestion and improving overall performance.

- **Reliability and Redundancy**

Bus topology offers low reliability due to its dependence on a single backbone cable, where failure can disrupt the entire network. Mesh topology, however, provides high reliability through redundancy, as multiple links ensure continuous communication even if some connections fail.

- **Cost Efficiency**

Bus topology is highly cost-efficient due to minimal cabling and no need for complex hardware, making it suitable for small or temporary networks. In comparison, mesh topology is very expensive because of extensive cabling and the requirement for multiple network interfaces, making it suitable only for critical applications where reliability is a priority.

4. Hybrid topology offers higher scalability than star topology by combining multiple network designs (like star, mesh, or bus). This allows easy expansion without disrupting current setups. In contrast, star topology is limited by the physical port capacity of its central hub or switch.

Key Differences in Scalability

Dependency on Central Device

- Star topology depends entirely on a central hub or switch for communication between nodes. While this makes management easier, it also creates a limitation. If the central device reaches its maximum port capacity, adding more devices becomes difficult without upgrading or adding another switch. Hybrid topology reduces this limitation by distributing network control across multiple segments, thereby avoiding complete dependence on a single device.

Performance and Traffic Management

- Hybrid topology provides better performance in large networks because traffic can be distributed across different segments. This reduces congestion and improves overall network efficiency. In contrast, star topology may experience performance issues if too many devices are connected to a single switch, leading to increased data traffic and possible delays.

Future upgradability

Hybrid topology is highly suitable for future expansion because it allows integration of new technologies and network structures without redesigning the entire system. Organizations can upgrade specific segments without affecting the rest of the network. In contrast, star topology may require hardware upgrades or restructuring when expanding beyond its capacity.

Overall, while both hybrid and star topologies support scalability, hybrid topology provides greater flexibility, better performance, and improved reliability for large-scale networks. Star topology, however, remains a simpler and cost-effective solution for smaller networks with limited expansion needs.

4. Maintenance Comparison of Star, Mesh, Bus, and Hybrid Topologies

Ease of Maintenance in Star Topology

Star topology is relatively easy to maintain because each device is connected to a central hub or switch. This centralized structure allows network administrators to quickly identify and isolate faults to a specific node or cable. If one connection fails, it does not affect the rest of the network, making troubleshooting faster and more efficient.

Maintenance Challenges in Mesh Topology

Mesh topology, although highly reliable, is difficult to maintain due to its complex structure. Each node is connected to multiple other nodes, resulting in a large number of links. This makes monitoring, managing, and troubleshooting more time-consuming. Identifying the exact point of failure among many connections can be challenging and requires skilled administration.

Maintenance Issues in Bus Topology

Bus topology is harder to maintain compared to star topology because all devices share a single backbone cable. If there is a fault in this main cable, the entire network can be disrupted. Additionally, locating the exact point of failure along the backbone can be difficult and time-consuming, especially in larger networks.

Maintenance in Hybrid Topology

Hybrid topology has moderate maintenance complexity. While it benefits from flexibility and can isolate faults within specific segments, managing a network that combines multiple topologies requires careful monitoring and expertise. Network administrators must understand and handle different configurations, which can increase the complexity of maintenance tasks.

Fault Isolation and Recovery

In terms of fault isolation, star topology performs well due to its centralized design. Mesh topology provides alternative paths for data, allowing communication to continue even during failures, but identifying issues is complex. Bus topology has poor fault isolation, as a single issue can affect the entire network. Hybrid topology offers better fault isolation than bus by dividing the network into segments.

Overall, star topology is the easiest to maintain, mesh topology is the most complex, bus topology presents challenges due to backbone dependency, and hybrid topology falls in between, offering a balance between flexibility and maintenance complexity.

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